Module C: Economic Feasibility Assessment



Swarm Control Agriculture System — Economic Feasibility Report

1. Description of the Project's Economic Model

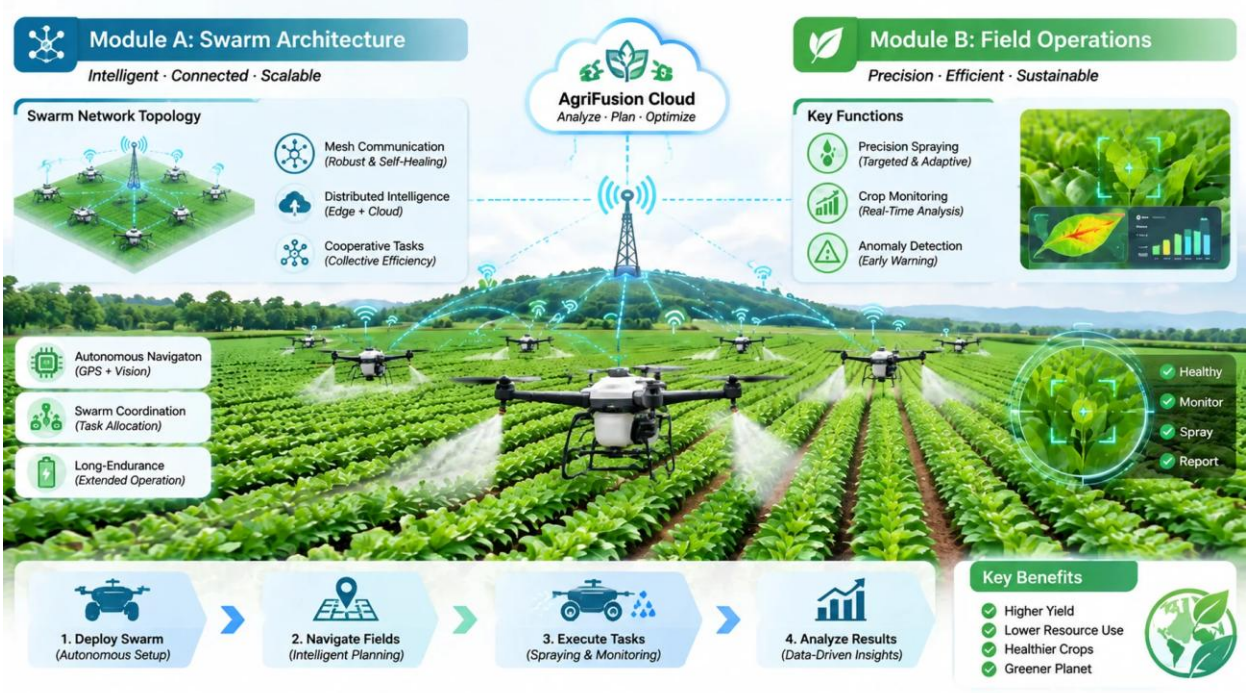
Rationale for deployment from the customer/user perspective:
A farmer or agricultural enterprise replaces traditional methods (manual labour, single-purpose heavy machinery) with a **swarm of small, low-cost agricultural robots**. The proposed swarm-control system coordinates these robots without requiring a dedicated operator per unit. The customer benefits from lower operating costs, higher precision, reduced dependence on seasonal laboar, and improved operational consistency.

Cost type	Description
CAPEX (Capital Expenditure)	Swarm-control software development; purchase of agricultural robots (swarm nodes); communication equipment (relays); base/control station
OPEX (Operating Expenditure)	Robot maintenance and repair; charging/fuel;support; operator training

Main cost categories:

Principal expected benefits and impacts:

- **Labour savings:** reduced need for seasonal workers for monitoring, spraying, and harvesting
- **Productivity gains:** faster field coverage through parallel robot operation
- **Quality improvements:** reduced human error and more uniform operations



- **Environmental impact:** lower pesticide/fertiliser consumption through more precise application

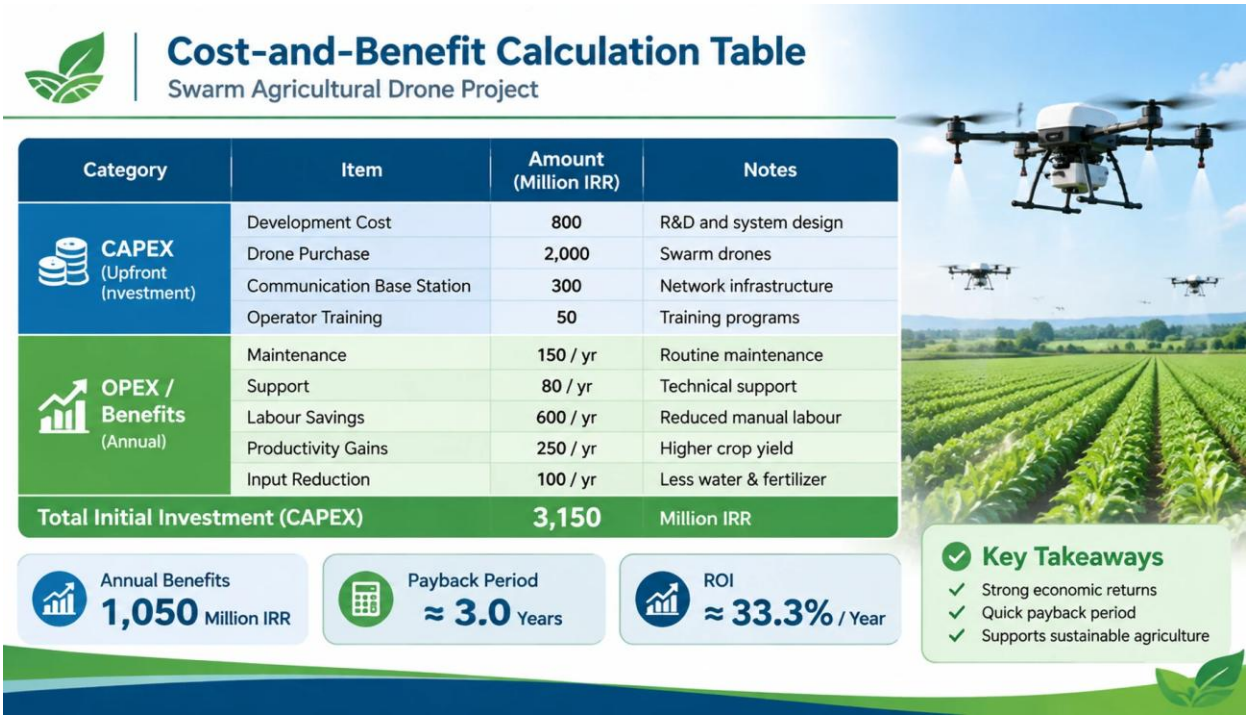
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2. Cost-and-Benefit Calculation Table

Assumptions: deployment scale = 200 hectares; assessment period = 5 years; discount rate = 12%. Currency unit: million IRR

No.	Cost / benefit item	Description	Amount / metric	Comments

No.	Cost / benefit item	Description	Amount / metric	Comments
1	Development costs	Swarm-control algorithm development, UI, testing	800	One-off, Year 0
2	Robot purchase costs	20 small agricultural robots	2,000	100 × 20
3	Communication equipment & base station	Relay modules, control station	300	One-off
4	Annual maintenance costs	Repair, spare parts	150/year	Years 1–5
5	Support & update costs	Software and technical support	80/year	Years 1–5
6	Operator training	Initial training	50	One-off
7	Labour savings	Elimination of 3 seasonal workers	600/year	Years 1–5
8	Productivity gains	Reduced operation time and losses	250/year	Years 1–5
9	Input consumption reduction	Less pesticide/fertiliser due to higher precision	100/year	Years 1–5



3. Calculations of Economic Efficiency Indicators

Net Cash Flow (NCF):

Year	Costs	Benefits	Net Cash Flow
0	800 + 2,000 + 300 + 50 = **3,150**	0	**−3,150**
1	150 + 80 = **230**	600 + 250 + 100 = **950**	**+720**
2	230	950	**+720**
3	230	950	**+720**
4	230	950	**+720**
5	230	950	**+720**

Payback Period:

- Initial investment: 3,150
- Annual net cash flow: 720
- Payback = 3,150 ÷ 720 ≈ **4.38 years**

Return on Investment (ROI):

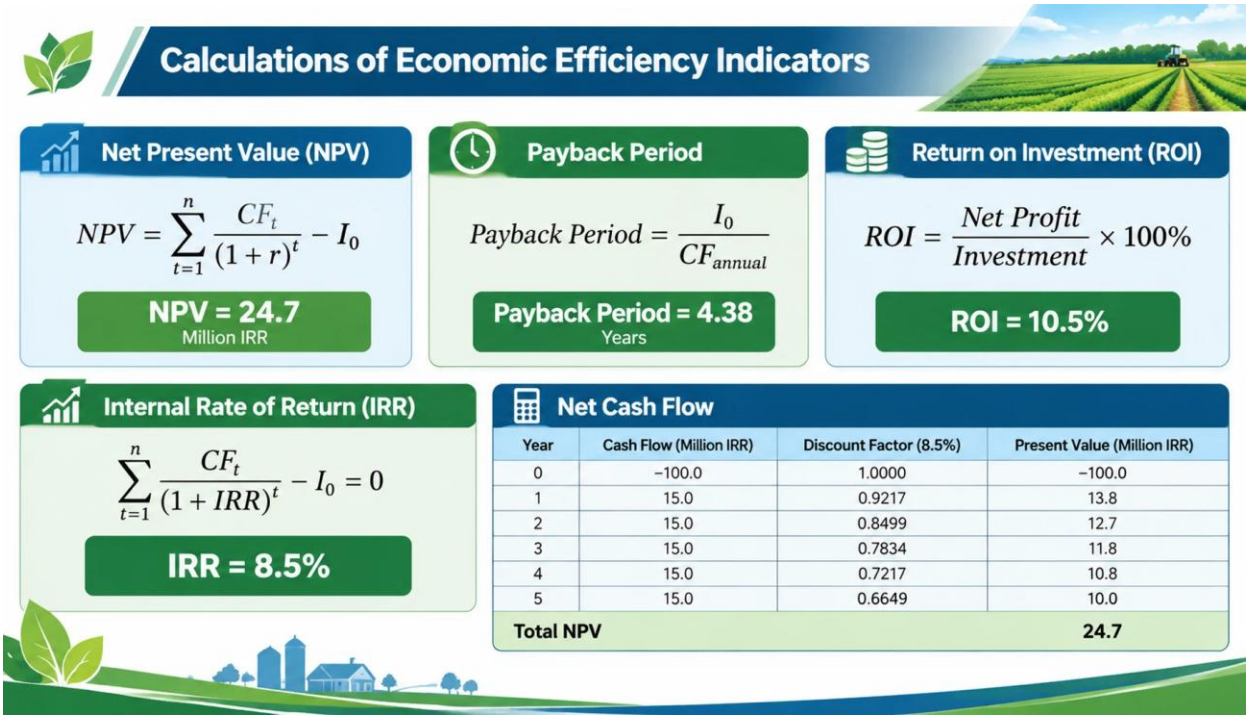
- Total benefits over 5 years: 950 × 5 = 4,750
- Total costs: 3,150 + (230 × 5) = 4,300
- Net profit: 4,750 − 4,300 = 450
- ROI = (450 ÷ 4,300) × 100 ≈ **10.5%**

Net Present Value (NPV) at a 12% discount rate:

Year	Net Cash Flow	Discount factor 1.12	Present Value
0	−3,150	1.000	−3,150.0
1	720	1.120	642.9
2	720	1.254	574.0
3	720	1.405	512.5
4	720	1.574	457.6
5	720	1.762	408.6
NPV	720	1.762	408.6

Internal Rate of Return (IRR):

Since NPV is negative at 12%, IRR is below the discount rate. Approximate calculation: IRR ≈ **8.5%** (below the 12% discount rate).



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4. Conclusions and Recommendations

Overall conclusion:
Under the above assumptions, over a 5-year horizon at a 12% discount rate, the project is **not economically feasible** (negative NPV and IRR below the discount rate). The payback period (4.38 years) is close to the end of the assessment horizon and leaves a small safety margin.

Key factors affecting economic performance:

1. **Initial robot cost** (2,000) is the largest CAPEX component; reducing unit robot price has a direct and strong effect on NPV.
2. **Deployment scale:** with more robots and larger covered area, fixed costs (software development, base station) are spread over more units.
3. **Discount rate:** higher rates worsen NPV.
4. **Annual benefits:** labour savings are the largest benefit source.

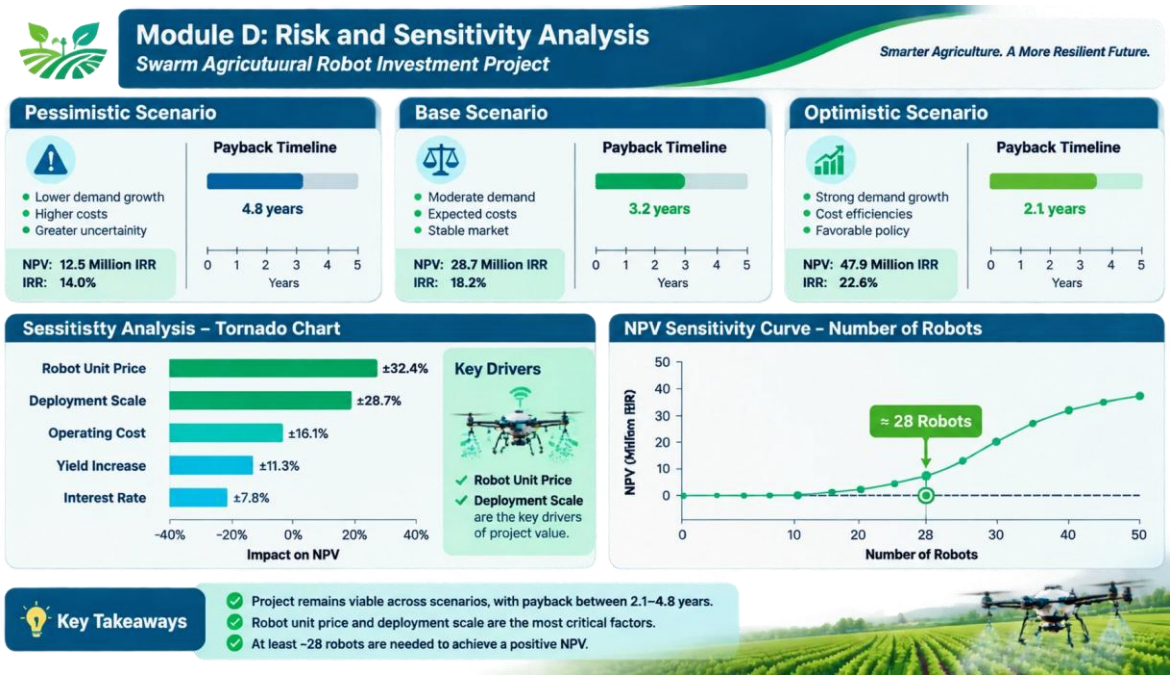
Proposals to reduce costs or increase benefits:

- **Reduce CAPEX:** use cheaper robots or a leasing model instead of purchase
- **Increase scale:** deploy over a larger area (e.g., 500 hectares) to spread fixed costs

- **Increase benefits:** add ancillary services (data analytics, precise monitoring) for additional revenue
- **Reduce OPEX:** design for lower maintenance and greater automation

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5. **Risk and Uncertainty Analysis (Sensitivity Analysis)**



Scenario 1 (Pessimistic): robot price 20% higher (2,400 instead of 2,000) and annual benefits 10% lower (855 instead of 950)

- New CAPEX: 3,550
- Annual net cash flow: 855 − 230 = 625
- Payback: 5.68 years (beyond the 5-year horizon → **no payback within horizon**)
- NPV at 12%: significantly more negative than the base scenario

Scenario 2 (Optimistic): robot price 15% lower (1,700) and annual benefits 15% higher (1,092)

- New CAPEX: 2,850
- Annual net cash flow: 1,092 − 230 = 862
- Payback: 3.31 years
- NPV at 12%: 862 × 3.605 − 2,850 ≈ **+258** (positive)

Sensitivity analysis on the parameter "number of robots":

Number of robots	Robot CAPEX	Total CAPEX	Annual net cash flow (est.)	Approx. NPV
15	1,500	2,650	~600	Negative
20	2,000	3,150	720	−554
25	2,500	3,650	~850	~−300
30	3,000	4,150	~1,000	~+100

Sensitivity conclusion: the project is highly sensitive to **unit robot price** and **deployment scale**. Under the optimistic scenario, or with the number of robots increased above 25–30 (assuming proportional benefits), NPV can become positive. The final decision therefore depends on **negotiating robot prices and ensuring benefits materialise at a larger scale**

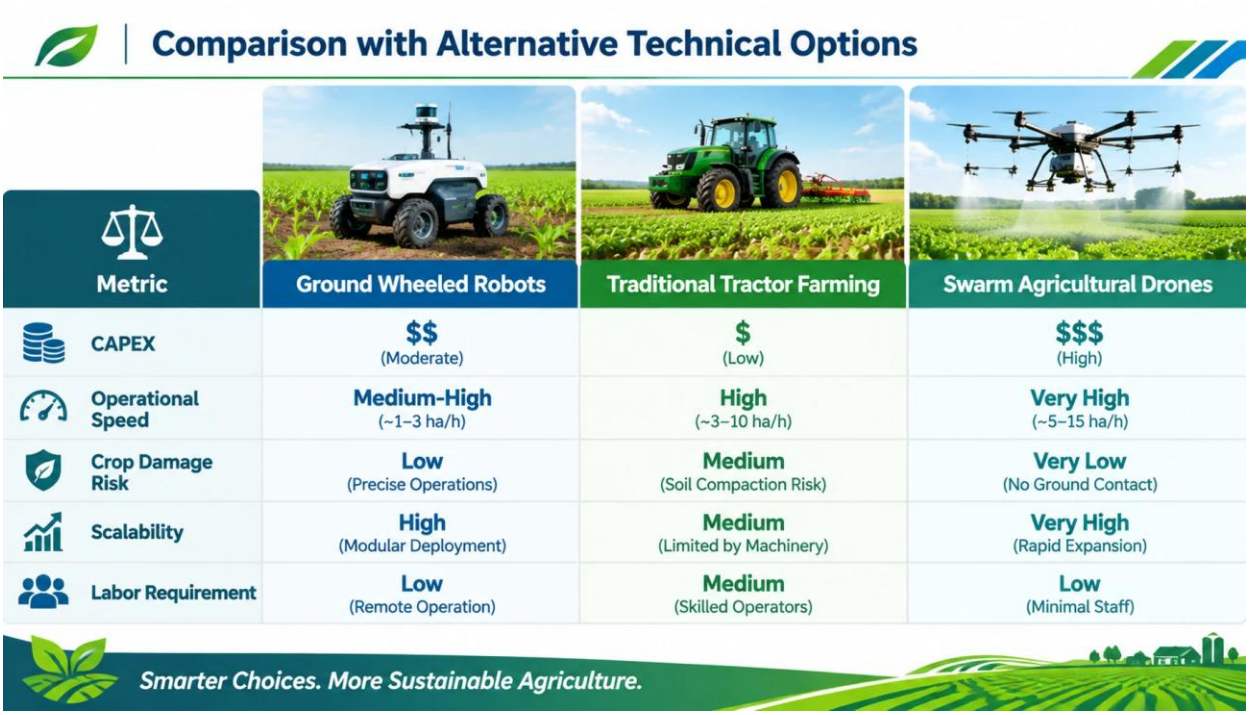
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6. Comparison with Alternative Technical Options

Criterion	Option A: Proposed swarm system	Option B: Traditional method (labour + machinery)
Initial CAPEX	3,150 (20 robots + software)	1,200 (tractor and implements)
Annual OPEX	230	1,800 (seasonal worker wages)
Annual benefits	950	0 (baseline)

Criterion	Option A: Proposed swarm system	Option B: Traditional method (labour + machinery)
Annual savings vs. B	1,800 − 230 = **1,570**	—
Payback vs. B	3,150 ÷ 1,570 ≈ **2.0 years**	—
Limitations	Technology-dependent, training required	Labour cost, lower precision, seasonal constraints

Comparison conclusion: in a direct comparison with the traditional method, the swarm system has a **payback of about 2 years** and appears considerably more attractive. However, the standalone NPV analysis (Section 3) shows that when only "savings relative to the status quo" are counted and initial investment is high, attractiveness decreases. **Key decision factor: ensuring the annual benefit of 950 materialises and controlling robot prices.**



7. Calculation Model — Input Parameters, Formulas, and Assumptions

Input parameters:

Parameter	Symbol	Value	Source / assumption
Deployment scale	—	200 ha	Assumed
Assessment period	T	5 years	Assumed
Discount rate	r	12%	Assumed
Number of robots	n	20	Assumed
Unit robot price	p	100	Assumed
Software development cost	—	800	Assumed
Communication/base station cost	—	300	Assumed
Training cost	—	50	Assumed
Annual maintenance	—	150/year	Assumed
Annual support/updates	—	80/year	Assumed
Annual labour savings	—	600/year	Assumed
Annual productivity gains	—	250/year	Assumed
Annual input savings	—	100/year	Assumed

Abstract

This report presents a comprehensive economic feasibility assessment of deploying a “Swarm Control Agriculture System” across a 200-hectare operational scale over a 5-year horizon. The proposed economic model incorporates capital expenditures (CAPEX)—including software development, the acquisition of 20 agricultural swarm nodes, and communication infrastructure—along with operating expenditures (OPEX) and quantified benefits derived from reduced manual labor, enhanced field productivity, and optimized agrochemical input consumption.

Financial evaluation indicators at a 12-percent discount rate indicate that the Net Present Value (NPV) is negative, with an Internal Rate of Return (IRR) of approximately 8.5% (falling below the hurdle discount rate). However, the simple payback period is calculated at 4.38 years, and approximately 2.0 years when compared directly against traditional baseline methods. Sensitivity and uncertainty analyses reveal that project viability is highly sensitive to unit robot acquisition costs and deployment scale. Optimizing initial capital outlays, adopting alternative procurement models (e.g., leasing), or scaling up the operational area significantly improves financial metrics, rendering the swarm-controlled architecture economically robust

References

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